
1. Almost Bingo

Program Name: Almost.java Input File: almost.dat

Judges Input File

```
4
1 16 31 46 61
2 17 32 47 62
3 18 FS 48 63
4 19 34 49 64
5 20 35 50 65
B12 I18 G48 x 068 062 G47 G50 B9 065 I17 B1 x
1 19 35 47 63
12 21 41 46 73
7 16 FS 59 61
14 29 34 55 70
2 30 40 60 68
I21 B14 I16 B2 B9 N34 I19 N45 G55 x 074 G46 063 x I29 B1 x
7 20 35 50 61
9 21 40 54 66
3 22 FS 57 65
8 23 38 60 75
12 24 45 48 70
I22 x N38 x B3 x N40 x B1 x G57 x 070 x B9 x 065 x
7 20 35 50 61
9 21 40 54 66
3 22 FS 57 65
8 23 38 60 75
12 24 45 48 70
I21 N40 G57 x G50 G54 G51 N45 x B7 G48 B8 I23 x B9 x 070 N35 061 x
```

Judges Output to Screen

```
3 2
\ 1

4 2 I 2 \ 2
/ 1
/ BINGO

3 3
3 3 N 3
3 2
3 2 N 2
3 2 N 2
3 1
3 1
3 1
3 BINGO

2 3 3 3 N 3 \ 3
2 2 N 2 G 2
G 1
2 1 G 1
1 1 2 1 N 1 G 1 \ 1 / 1
```

2. Butterflies

Program Name: Butterflies.java

Input File: butterflies.dat

Judges Input File

18
MONARCH, ORANGE
CLOUDED SULPHUR, YELLOW
LYSCIDE SULPHUR, YELLOW
PAINTED LADY, ORANGE
GIANT SWALLOWTAIL, BLACK
CLOUDLESS SULPHUR, YELLOW
BLACK SWALLOWTAIL, BLACK
TIGER SWALLOWTAIL, YELLOW
GULF FRITILLARY, ORANGE
SOUTHERN DOGFACE, YELLOW
HACKBERRY EMPEROR, YELLOW
QUEEN, ORANGE
CABBAGE WHITE, WHITE
LONG-TAILED SKIPPER, BLACK
QUESTION MARK, ORANGE
LEAST SKIPPER, YELLOW
PEARL CRESCENT, ORANGE
GREAT SOUTHERN WHITE, WHITE
4
YELLOW
BLACK
ORANGE
WHITE

Judges Output to Screen

YELLOW:
CLOUDED SULPHUR
CLOUDLESS SULPHUR
HACKBERRY EMPEROR
LEAST SKIPPER
LYSCIDE SULPHUR
SOUTHERN DOGFACE
TIGER SWALLOWTAIL

BLACK:
BLACK SWALLOWTAIL
GIANT SWALLOWTAIL
LONG-TAILED SKIPPER

ORANGE:
GULF FRITILLARY
MONARCH
PAINTED LADY
PEARL CRESCENT
QUEEN
QUESTION MARK

WHITE:
CABBAGE WHITE
GREAT SOUTHERN WHITE

3. Credit Card

Program Name: CreditCard.java

Input File: creditcard.dat

Judges Input File

8

1234567891234567

4552720412345677

9087654321135793

5490123456789128

5588320123456789

4408041234567893

4417123456789113

4321567890621945

Judges Output to Screen

INVALID

VALID

INVALID

VALID

INVALID

VALID

VALID

INVALID

4. DNA

Program Name: DNA.java **Input File:** dna.dat

Judges Input File

```
4
ATGGCATAAGTAT
TGCAGCTGCATCAGGAT
ACTG
TGCA
AAAAAAAAATTTTTTTTTTTTTTTT
CCCCCCCCCCCCGGGGGGGGGGGGGGGG
TACGTACGTACGATCG
GCTAGCTATCGCTA
```

Judges Output to Screen

```
4 GCAT
2 TG
1
4 ATCG
```

5. DVDs

Program Name: DVDs.java

Input File: dvds.dat

Judges Input File

Transformers, Blades of Glory, Disturbia, 300, Spider-Man 3 (2), Ghost Rider,
TMNT, Star Trek Generations (2), My Family: Season 2 (2), Norbit, Because I
Said So, Eragon, Walk the Line, The Holiday, Grounded for Life Season 5 (2)
5
Norbit, Speed, My Family: Season 2 (2), Ghost Rider, King Kong
Eragon
Blades of Glory, Because I Said So, The Holiday, Grounded for Life Season 5
(2), Crash, Speed, Shark Tale, Curious George, Ice Age
King Kong
Because I Said So, Walk the Line, Fool's Gold, The Holiday, Bucket List, 16
Blocks, Transformers SE (2), Aeon Flux
Fool's Gold
King Kong, Ghost Rider, Speed
Grounded for Life Season 5 (2)
Crash
300

Judges Output to Screen

BETA 5
BETA 1
ALPHA 6
BETA 7
GAMMA 5

6. Fibonacci Base

Program Name: FibonacciBase.java

Input File: fibonaccibase.dat

Judges Input File

16
29
53
92
999
846
745
377

Judges Output to Screen

16 = 13 + 3
29 = 21 + 8
53 = 34 + 13 + 5 + 1
92 = 89 + 3
999 = 987 + 8 + 3 + 1
846 = 610 + 233 + 3
745 = 610 + 89 + 34 + 8 + 3 + 1
377 = 377

7. Magic Square

Program Name: Magic.java

Input File: magic.dat

Judges Input File

```
7
8 11 23 1
22 2 7 12
3 25 9 6
10 5 4 24
8 11 23 1
22 2 7 12
3 25 9 6
10 5 4 19
22 1 12 7
11 8 21 2
5 10 3 24
4 23 6 9
10 5 4 19
22 1 12 7
11 8 20 2
5 10 3 24
7 2 16 9
12 13 3 6
1 8 10 15
14 11 5 4
34 1 12 7
11 8 33 2
5 10 3 36
4 35 6 9
80 1 12 7
11 8 79 2
5 10 3 82
4 81 6 9
```

Judges Output to Screen

```
43 MAGIC
NOT MAGIC
42 MAGIC
NOT MAGIC
34 MAGIC
54 MAGIC
100 MAGIC
```

8. Making a Difference

Program Name: Difference.java

Input File: difference.dat

Judges Input File

78.3
92.1
78.4
62.4
93.4
90.0
99.9
90.8
88.2
87.3
22.0
41.6
87.5
92.1

Judges Output to Screen

81

9. Paint

Program Name: Paint.java

Input File: paint.dat

Judges Input File

```
3
9
XXXXXXXX==XXXXXXXXXXXXXXXXXX
X      X      X
X      *      X      X
X      -      X
X      X      X
XXXXXXXX==XXXXXXXX - - XXXXXXXXXXXX
X      X      -      X
X      X      X      =
XXXXXXXXXXXXXXXXXXXX==XXXXX==XX
14
XXXXXXXXXXXXXXXXXXXX
X      =
X      =
X      =
X      X
XXXXX - XXXXX - XXXXXXXX==XXXXX
X      X      X      X
X      X      X      X
X      X      XXXXX - XXXXX
=      -      X      X
X      X      -      *      X
X      X      X      X
X      X      X      X
XXXXXXXXXXXXXXXXXXXX==XXXXX
12
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
X      X
X      X
X      *      XXXXXXXXXXXXX      X
X      X      X      =
=      =      =
=      X      X      X
XXXXXX==XXXXXX==XXXXX      X
X      X      X      X
X      -      -      X
X      X      X      X
XXXXX==XXXXXX==XXXXXX==XXXXX
```

Judges Output to Screen

```
XXXXXXXX==XXXXXXXXXXXXXXXXX
XWWWWWWWWWWXWWWWWWWWWWX
XWCCCCCCCCWXCWCCCCCCCCW
XWCCCCCCCCCCCCCCCCCCCCW
XWWWWWWWWWWXWWWWCWWWWX
XXXXXXXX==XXXXXXXXCCXXXXXXXXX
X          XWWWWCWWWWCCWWWWX
X          XWWWWWWWWWWXWWWW=
XXXXXXXXXXXXXXXXXX==XXXXX==XX
18
```

```
XXXXXXXXXXXXXXXXXX
XWWWWWWWWWWWW=
XWCCCCCCCCCCCCCW=
XWCCCCCCCCCCCCCW=
XWWWWCWWWWCWWWWX
XXXXXCXXXXCXXXXXX==XXXXX
XWWWWCWWXWWCWWXWWWWWWWWX
XWCCCCCWXCWCCCCWXCWWWCWWWWX
XWCCCCCWXCWCCCCWXXXXCXXXXX
=WCCCCCCCCCCCCCWXCWWWCWWWWX
XWCCCCCWXCWCCCCCCCCCCCCCW
XWCCCCCWCCCCCWXCWCCCCCW
XWWWWWWWWWWWWXWWWWWWWWX
XXXXXXXXXXXXXXXXXX==XXXXX
32
```

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XWWWWWWWWWWWWWWWWWWWWWWWWX
XWCCCCCCCCWXXXXXXXXXXCWWX
XWCCCCCCCCW          XWCCCCW=
=WCCCCCCCCW=          =WCCCCW=
=WWWWWWWWX          XWCCCCW
XXXXXXXX==XXXXXX==XXXXXWCCCCW
XWWWWWWWWXWWWWWWWWXWCCCCW
XWCCCCCCCCCCCCCCCCCCCCCCCCW
XWWWWWWWWXWWWWWWWWXWWWWWWX
XXXXX==XXXXXX==XXXXXX==XXXX
31
```

10. Pyramid Scheme

Program Name: Pyramid.java

Input File: pyramid.dat

Judges Input File

```
12
3 4
1 2 3 1
1 2 5 1
1 1 1 1
1 1
12
5 6
11 12 13 14 13 2
13 15 16 16 15 2
11 15 22 22 15 1
11 17 17 15 16 2
11 12 13 14 10 5
2 5
5 12 5 12 4
3 22 9 37 9
5 1
15
14
13
12
10
1 5
12 23 12 45 100
5 5
1 2 3 4 5
1 11 11 11 5
8 12 45 12 10
9 40 40 40 10
10 10 10 10 10
7 7
1 2 3 4 5 1 1
1 11 11 11 12 12 5
8 12 45 45 100 10 4
9 17 40 100 40 12 3
8 12 45 45 99 10 4
1 11 11 11 12 12 5
1 2 3 4 5 1 1
3 6
1 1 1 1 1 1
1 20 10 5 10 1
1 1 1 1 5 1
3 7
1 1 1 1 1 1 1
1 20 10 12 10 20 1
1 1 1 1 5 1 8
8 4
5 6 7 5
3 10 10 5
```

9 12 12 9
1 10 10 9
4 20 20 9
9 30 30 9
4 10 10 9
4 5 7 1
5 5
1 1 1 1 1
1 5 5 5 1
1 5 1 5 1
1 5 5 5 1
1 1 1 1 1

Judges Output to Screen

SCRAP IT
BUILD IT
BUILD IT
BUILD IT
BUILD IT
BUILD IT
BUILD IT
SCRAP IT
SCRAP IT
BUILD IT
BUILD IT
SCRAP IT

11. Rose Petals

Program Name: Roses.java

Input File: roses.dat

Judges Input File

```
4
ALAN, BRYAN, CHAD, DUKE, ERIC, FRED
100 234 564 234 1231
LUCAS, MARK, NOEL, PERRY, QUINCEY, REX, SAM
250 231 198 42 54 99
GUY, HANK
601
JACK, KIT, LEN, MATT
601 402 1
```

Judges Output to Screen

```
LOSER DUKE
LOSER BRYAN
LOSER ALAN
LOSER FRED
LOSER CHAD
WINNER ERIC

LOSER QUINCEY
LOSER LUCAS
LOSER PERRY
LOSER SAM
LOSER REX
LOSER MARK
WINNER NOEL

LOSER GUY
WINNER HANK

LOSER JACK
LOSER MATT
LOSER KIT
WINNER LEN
```

12. Time's Up

Program Name: Time.java

Input File: none

Judges Input File

None

Judges Output to Screen

[illegible]